

Unit 1

1. Program to Demonstrate Basic I/O Functions Using Different Data Types

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int age;
```

```
    float height;
```

```
    double salary;
```

```
    char grade;
```

```
    printf("Enter age: ");
```

```
    scanf("%d", &age);
```

```
    printf("Enter height: ");
```

```
    scanf("%f", &height);
```

```
    printf("Enter salary: ");
```

```
    scanf("%lf", &salary);
```

```
    printf("Enter grade: ");
```

```
    scanf(" %c", &grade);
```

```
    printf("\n--- Details ---\n");
```

```
    printf("Age = %d\n", age);
```

```
    printf("Height = %.2f\n", height);
```

```
printf("Salary = %.2lf\n", salary);  
printf("Grade = %c\n", grade);  
  
return 0;  
}
```

2. Program to Calculate the Area of a Circle

```
#include <stdio.h>
```

```
int main()  
{  
    float radius, area;  
    const float PI = 3.14159;  
  
    printf("Enter radius of circle: ");  
    scanf("%f", &radius);  
  
    area = PI * radius * radius;  
  
    printf("Area of circle = %.2f\n", area);  
  
    return 0;  
}
```

3. Program to Calculate the Average of Three Numbers

```
#include <stdio.h>
```

```
int main()
```

```
{  
    float a, b, c, average;  
  
    printf("Enter three numbers: ");  
    scanf("%f %f %f", &a, &b, &c);  
  
    average = (a + b + c) / 3;  
  
    printf("Average = %.2f\n", average);  
  
    return 0;  
}
```

4. Program to Swap Two Numbers Using a Temporary Variable

```
#include <stdio.h>
```

```
int main()  
{  
    int a, b, temp;  
  
    printf("Enter two numbers: ");  
    scanf("%d %d", &a, &b);  
  
    printf("Before swapping: a = %d, b = %d\n", a, b);  
  
    temp = a;  
    a = b;
```

```
b = temp;

printf("After swapping: a = %d, b = %d\n", a, b);

return 0;
}
```

5. Program to Demonstrate Single-Line and Multi-Line Comments

```
#include <stdio.h>

int main()
{
    // This is a single-line comment.
    printf("Hello, World!\n");

    /*
       This is a multi-line comment.
       It can contain multiple lines.
    */

    printf("This program demonstrates comments in C.\n");

    return 0;
}
```